Core Evidence 4: Exploratory Data Analysis (EDA) Insight Log

IT3091 Machine Learning — Industry Explorer Track

Group ID: `2026-AI-36`

Project: ChefCast AI — Restaurant Revenue & Product Demand Forecasting

Version: Final Submission — October 2026

> *This log documents every significant exploratory finding that directly influenced a downstream preprocessing, feature engineering, or modeling decision. Every insight is connected to a specific decision or architectural choice — EDA was not conducted as a passive description exercise but as active evidence collection for defensible ML design.*

1. Raw Dataset Characterization

Property	Value	Analytical Implication
Total raw transaction rows	127,093 line items	Large enough for time-series ML with meaningful signal
Unique customer invoices	45,947 tickets	Confirms genuine retail trading volume
Unique catalog items	274 distinct items	Catalog breadth necessitates principled product scoping (DEC-03)
Raw date span	2009-01-05 to 2026-09-07	17-year span flags data quality anomalies before commercial launch
Authentic operational window	2024-03-27 to 2026-09-07	895 calendar days of genuine restaurant trading
Retained after cleaning	125,912 transactions (99.1%)	Minimal data loss; cleaning was targeted, not sweeping

2. Data Quality Audit: Four Anomaly Classes

Insight 2.1 — Pre-Commissioning System Records (147 rows)

What was found: 147 transactions are dated 2009 (117 rows), 2011 (15 rows), and 2020 (15 rows). These predate any plausible Sri Lankan restaurant operating in 2024.

Evidence analysis:

- The 2009/2011 records cluster around January with round-number quantities (exactly 1.0, 2.0) at fixed prices — consistent with POS terminal factory test patterns.
- The 2020 records are isolated, suggesting either a temporary system reset or branch network clock sync errors during COVID-era downtime.

Decision consequence: All records before 2024-03-27 were filtered (DEC-04). Retaining them would introduce revenue "ghost spikes" in 2009 and 2020 that the model's lag features would propagate forward as false seasonality signals.

Insight 2.2 — Transaction Voids & Customer Refunds (931 rows)

What was found: 931 records carry negative or zero quantities, including extreme entries as negative as ***-1{,}111{,}111{,}000***.

Evidence analysis:

- Negative quantity patterns in hospitality POS systems are a standard implementation practice for credit/void entries: the original sale is recorded as +Qty, and the reversal is recorded as –Qty against the same invoice.
- The extreme outlier (***-1,111,111,000*** units) is clearly a software field-overflow artefact, not a genuine return.

Decision consequence: Filtered entirely (DEC-04). Converting to absolute values would double-count demand — a customer who returns one roll does not add a positive unit to genuine demand.

Modeling implication: The cleaned dataset represents net-consumed quantities only — the economically correct measure of actual demand that a kitchen must prepare for.

Insight 2.3 — Promotional Price Adjustments (72 rows)

What was found: 72 records carry negative unit prices as extreme as ***-19{,}602{,}000*** LKR.

Evidence analysis:

- POS systems in South Asian food & beverage environments frequently implement promotional campaigns as negative-price line items applied to the total invoice (e.g., a **`DISCOUNT -500 LKR`** entry).
- These are not actual food items — they are billing arithmetic adjustments appended as catalog-entry line items.

Decision consequence: Filtered from the revenue aggregation (DEC-04). Including them would cause cumulative daily revenue to appear drastically reduced or even negative on promotion-heavy days, corrupting the revenue target variable.

Insight 2.4 — Extreme Price Anomalies (52 rows)

What was found: Multiple records with **`UnitPrice`** exceeding ***2.41 × 10^{11}*** LKR per unit.

Evidence analysis:

- Sri Lankan restaurant dish prices realistically range from LKR 50 (a cup of tea) to LKR 1,500 (a full Chicken Rice). A price of ***241,000,000,000*** LKR for a single dish is 161 million times the GDP of Sri Lanka.
- These entries are consistent with numeric field-length overflow in legacy POS software (e.g., a barcode prefix accidentally populating the price field).

Decision consequence: Applied upper bound filter: ***{UnitPrice} \leq 20{,}000*** LKR and ***{Qty} \leq 300*** units (DEC-04). These bounds are generous enough to accommodate genuine premium catering orders while excluding clear data corruption.

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3. Revenue Concentration & Pareto Distribution Analysis

Insight 3.1 — 11 Core Items Generate 47.0% of Total Revenue

Finding: Despite 274 unique items in the catalog, the 11 selected fresh handmade dishes generate approximately **34,322,871 LKR (~47.0%)** of total cleaned store turnover — a classic Pareto (80/20-type) concentration.

Evidence: Revenue share was computed per item across the full operational window, then ranked descending. The cutoff was applied at items satisfying all 5 selection criteria simultaneously.

Decision consequence: Product selection was scientifically grounded (DEC-03), not arbitrary. Forecasting only these 11 items captures the majority of kitchen-relevant revenue while avoiding noise from sporadic or non-perishable catalog entries.

Insight 3.2 — Volume Hierarchy Within the 11 Items

Rank	Item	Total Units	Mean Daily Units	Demand Variability	Kitchen Risk
1	`LUNCH`	18,231.5	20.4	Moderate (consistent)	**HIGH** — Core lunch service
2	`BREAKFAST`	15,670.2	17.5	Moderate (consistent)	**HIGH** — Early morning batch
3	`ROLLS`	14,958.0	16.7	High (weekend peaks)	**HIGH** — Perishable short-eats
4	`TEA`	10,973.0	12.3	Low (stable)	**MEDIUM** — Continuous service
5	`CHICKEN CURRY`	10,460.0	11.7	High (festive spikes)	**VERY HIGH** — Batch cook, high cost
6	`PATIS`	4,757.0	5.3	High	**HIGH** — Handmade, short shelf-life
7	`WATALAPPAN`	4,566.0	5.1	Very High (intermittent)	**HIGH** — Steam-cooked, <6h shelf life
8	`CHICKEN RICE FULL`	3,594.0	4.0	High	**MEDIUM** — Wok-cooked on order
9	`DATE CAKE`	2,320.0	2.6	Very High (intermittent)	**HIGH** — Baked once per day
10	`SAMOSA`	1,851.0	2.1	Very High (intermittent)	**HIGH** — Handmade, deep-fried

11	`BUTTER CAKE`	1,838.0	2.1	Very High (intermittent)	**HIGH** — Baked once per day

Key modeling implication: Items ranked 8–11 exhibit highly intermittent demand with frequent zero-demand days. This is why MAPE is unsuitable (DEC-08) — a single zero-demand day on `DATE CAKE` collapses the MAPE metric for that fold. WAPE handles these seamlessly.

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4. Temporal Demand Pattern Analysis

Insight 4.1 — Strong 7-Day Weekly Cyclicality

Finding: Revenue and demand exhibit a clear 7-day repeating pattern:

- **Peak days:** Friday, Saturday, Sunday — 25%–40% above weekday baselines. Weekend dinners and family short-eat orders dominate.
- **Trough days:** Tuesday and Wednesday — 15%–25% below the weekly average, consistent with lower weekday footfall in casual-dine Sri Lankan restaurants.

Evidence: Computed 7-day rolling mean vs. individual daily observations. Day-of-week box plots show systematically non-overlapping interquartile ranges between weekday and weekend groups.

Decision consequence: Directly motivated the ***{lag}_7*** feature as the primary autoregressive predictor. The 7-Day Seasonal Naive baseline (***\hat{y}_t = y_{t-7}***) was selected as the industry benchmark because it perfectly captures this single dominant cycle.

Insight 4.2 — Monthly & Festival Calendar Seasonality

Finding: Three annual seasonal peaks are identifiable:

- **December (Christmas / Year-End):** +30%–60% revenue surge, driven by festive family dining and gift cake orders.
- **April (Sinhala & Tamil New Year):** Significant spike from traditional sweet and savory short-eat demand (`PATIS`, `SAMOSA`, `ROLLS`) coinciding with the Sri Lankan New Year.
- **August (Annual school holiday):** Sustained 20%–40% elevated demand across all dish categories.

Evidence: Monthly aggregates show systematic inter-month differences consistent with the Sri Lankan cultural calendar. The pattern is reproducible across both 2024 and 2025 data.

Decision consequence: Motivated `month`, `quarter`, and ***\sin/\cos({month})*** cyclical encodings as feature engineering components. The ***\sin/\cos*** encoding preserves continuity between December (month=12) and January (month=1), which a linear `month` integer would break.

Insight 4.3 — Zero-Demand Days and Calendar Ambiguity

Finding: The dataset contains 74 dates with zero total qualifying POS sales across all 11 items — spread across Poya (full-moon) public holidays and isolated weekdays.

Evidence: These dates cannot be definitively classified as:

- Confirmed restaurant closures (no external business hours data available), or
- POS terminal connectivity failures or offline operational days.

Decision consequence: Zero-filled reindexing was adopted (DEC-05) with explicit documentation that these are characterized as *"no qualifying POS sales recorded"* rather than confirmed closure days. The distinction matters because the ML model will learn to predict near-zero on these calendar positions — which is correct operationally (the restaurant should not prepare large batches on Poya days regardless of cause).

Insight 4.4 — Demand Intermittency in Low-Volume Dishes

Finding: Items `DATE CAKE`, `BUTTER CAKE`, `SAMOSA`, and `WATALAPPAN` exhibit heavy-tailed zero-inflated daily distributions — on 15%–30% of operating days, these items record zero or near-zero demand.

Statistical evidence: Mean/variance ratios (coefficient of variation) for these items exceed 1.5, indicating over-dispersed demand distributions well beyond a Poisson assumption.

Key modeling implication: This explains why Ridge Regression outperforms gradient boosting methods on Lens 2. LightGBM and XGBoost attempted to model the non-linear zero-spike patterns but overfit to idiosyncratic noise clusters. Ridge's L2 regularization smoothed over these intermittent zero-demand events to produce more generalizable predictions on held-out test data.

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5. Feature Correlation & Mutual Information Analysis

Insight 5.1 — Strongest Revenue Predictors

Feature	Pearson Correlation with Revenue	Domain Explanation
`Revenue_lag_7`	0.71	Same weekday last week — strongest single predictor
`Revenue_roll_mean_7`	0.68	7-day trailing average — smoothed weekly baseline
`Revenue_lag_1`	0.62	Previous day momentum
`Revenue_roll_mean_14`	0.61	Fortnightly demand baseline
`is_weekend`	0.38	Binary weekend indicator
`sin_month`	0.22	Seasonal festival calendar cycle

Implication: High correlations among lag features (multicollinearity) makes Ridge Regression's L2 regularization directly appropriate — it shrinks correlated coefficient magnitudes harmoniously without eliminating features.

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6. EDA Summary: How EDA Directly Shaped Model Architecture

EDA Finding	Decision Triggered	Impact on Model
4 distinct anomaly classes in raw data	DEC-04: Targeted cleaning	Prevented revenue target corruption
7-day weekly cyclicality dominates	DEC-06: lag_7 as primary feature	Captures strongest single predictive signal
Zero-demand Poya days ambiguity	DEC-05: Zero-fill reindexing	Preserved temporal continuity without data invention
Low-volume dish intermittency	DEC-08: WAPE over MAPE	Prevented infinite metric values on zero-demand days
Multicollinearity among lag features	DEC-09: Include Ridge Regression	L2 shrinkage handles correlated predictors gracefully
Festival calendar seasonal spikes	DEC-06: sin/cos month encodings	Preserves circular boundary between December and January